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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Logic Families

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

28	18	1987 – 2022	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	21 questions
Marks 2	6 questions
Marks 8	1 question
Answer key	at the end

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Marks 1 21 questions

Q1. GATE ECE 2022

Open solution

Select the correct statement(s) regarding CMOS implementation of NOT gates.

(A) Noise Margin High (NM_H) is always equal to the Noise Margin Low (NM_L), irrespective of the sizing of transistors.

(B) Dynamic power consumption during switching is zero.

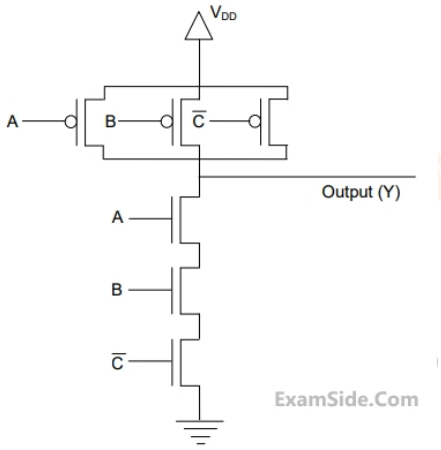
(C) For a logical high input under steady state, the nMOSFET is in the linear regime of operation.

(D) Mobility of electrons never influences the switching speed of the NOT gate.

Q2. GATE ECE 2014 Set 4

Open solution

The output (Y) of the circuit shown in the figure is



(A) $\overline{A}+\overline{B}+C$

(B) $A+\overline{B}.\overline{C}+A.\overline{C}$

(C) $\overline{A}+B+\overline{C}$

(D) $A.B.\overline{C}$

Q3. GATE ECE 2009

Open solution

The full forms of the abbreviations TTL and CMOS in reference to logic families are

(A) Triple Transistor Logic and chip metal oxide semiconductor.

(B) Tri-state Transistor Logic and chip metal oxide semiconductor.

(C) Transistor Transistor Logic and complementary metal oxide semiconductor.

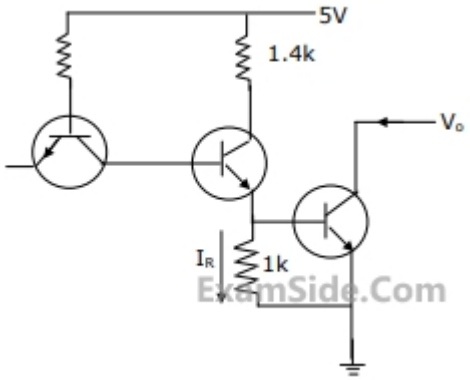
(D) Tri-state Transistor Transistor Logic and complementary metal oxide silicon.

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Q4. GATE ECE 2005

Open solution

The transistors used in a portion of the TTL gate shown in figure have β=100. the base-emitter voltage of is 0.7V for a transistor in active region and 0.75V for a transistor in saturation . If the sink current I=1mA and the output is at logic 0, then the current I_R I will be equal to

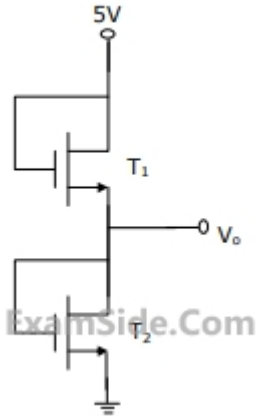


(A) 0.65 mA (B) 0.70 mA (C) 0.75 mA (D) 1.00 mA

Q5. GATE ECE 2005

Open solution

Both transistors T1 and T2 in figure have a threshold voltage of 1 Volt. The device parameters K_1 and K_2 of T_1 and T_2 are, respectively, 36 μA/V^2 and 9 μA/V^2. The output voltage V_0 is

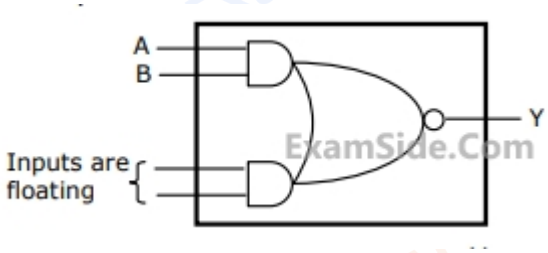


(A) 1V (B) 2V (C) 3V (D) 4V

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Q6. GATE ECE 2004

Figure shows the internal schematic of a TTL AND-OR-Invert (AOI) gate. For the inputs shown in Figure, the output Y is



(A) 0 (B) 1V (C) AB (D) $\overline{AB}$

Open solution

Q7. GATE ECE 2003

The output of the 74 series of TTL gates is taken from a BJT in

(A) Totem pole and common collector configuration

(B) either totem pole or open collector configuration

(C) common base configuration

(D) common collector configuration

Open solution

Q8. GATE ECE 1999

A Darlington Emitter follower circuit is sometimes used in the output stage of a TTL gate in order to

(A) increase its I_{OL} (B) Reduce its I_{OH}

(C) Increase its speed of operation. (D) Reduce power Dissipation.

Open solution

Q9. GATE ECE 1999

Commercially available ECL gates use two ground lines and one negative supply in order to

(A) reduce power dissipation

(B) increase fan-out

(C) reduce loading effect

(D) eliminate the effect of power line glitches or the biasing circuit.

Open solution

Q10. GATE ECE 1998

The noise margin of a TTL gate is about

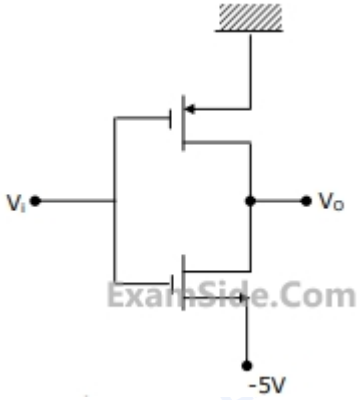
(A) 0.2V (B) 0.4V (C) 0.6V (D) 0.8V

Open solution

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Q11. GATE ECE 1998

The threshold voltage for each transistor in Fig.2.5, is 2V. For this circuit to work as an inverter, Vi must take the values



Open solution

(A) -5 V and 0 V (B) -5 V and 5 V (C) -0 V and 3 V (D) 3 V and 5 V

Q12. GATE ECE 1997

In standard TTL the 'totem pole' stage refers to

(A) the multi-emitteer input stage (B) the phase splitter

(C) the output buffer (D) open collector output stage

Open solution

Q13. GATE ECE 1997

The inverter 74AL SO4 has the following specifications:
$I_{OH\ \max} = -$ 0.4mA, I_{OL} max = 8mA, I_{IH} max = μA , $I_{IL\ \max}$ =0.1mA. The fan out based on the above will be

Open solution

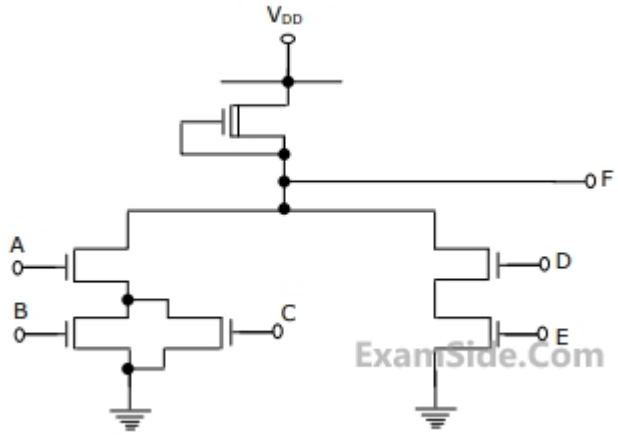
(A) 10 (B) 20 (C) 60 (D) 100

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Q14. GATE ECE 1997

Open solution

For the NMOS logic gate shown in figure, the logic function implemented is



(A) $\overline{ABCDE}$

(B) $(AB+\overline{C}).(\overline{D+E})$

(C) $\overline{A.(B+C)+D.E}$

(D) $(\overline{A+B}).C+\overline{D}.\overline{E}$

Q15. GATE ECE 1997

Open solution

The gate delay of an NMOS inverter is dominated by charge time rather than discharge time because

(A) the driver transistor has a larger threshold voltage than the load transistor.

(B) the driver transistor has larger leakage currents compared to the load transistor.

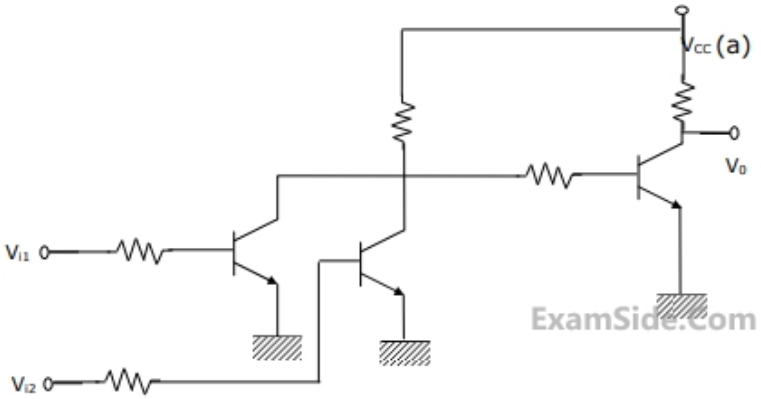
(C) the load transistor has a smaller W/L ratio compared to the driver transistor

(D) none of the above

Q16. GATE ECE 1992

Open solution

The figure shows the circuit of a gate in the Resistor Transistor Logic (RTL) family. The circuit represents a



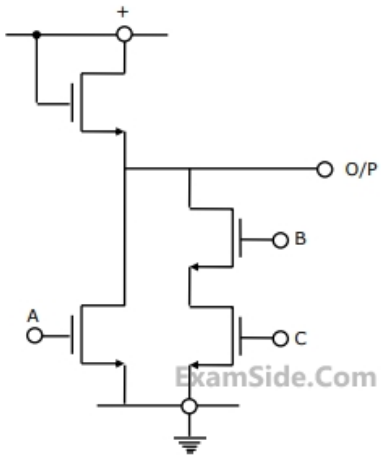
(A) NAND (B) AND (C) NOR (D) OR

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Q17. GATE ECE 1991

The CMOS equivalent of the following n MOS gate (figure) is _____ (draw the circuit).

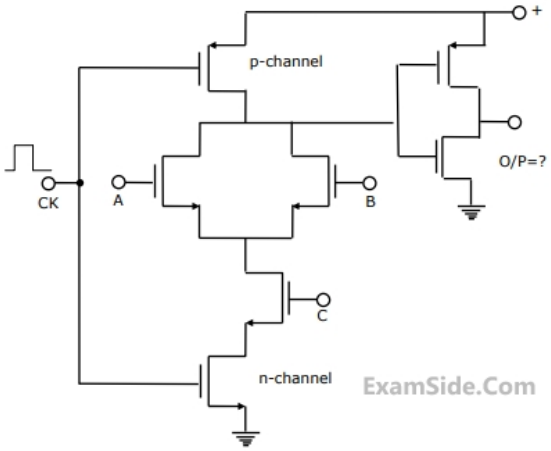
Open solution



Q18. GATE ECE 1991

In figure, the Boolean expression for the output in terms of inputs A, B and C when the clock 'CK' is high, is given by _____.

Open solution



Q19. GATE ECE 1989

Among the digital IC-families-ECL, TTL and CMOS:

(A) ECL has the least propagation delay
(B) TTL has the largest fan-out
(C) CMOS has the biggest noise margin
(D) TTL has the lowest power consumption

Open solution

Q20. GATE ECE 1989

A logic family has threshold voltage V_T= 2V, minimum guaranteed output high voltage V_{OH}= 4V, minimum accepted input high voltage V_{IH} = 3V, maximum guaranteed output low voltage V_{OL} = 1V, and maximum accepted input low voltage N V_{IL} =1.5V. Its noise margin is

(A) 2V
(B) 1V
(C) 1.5V
(D) 0.5V

Open solution

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Q21. GATE ECE 1987

Open solution

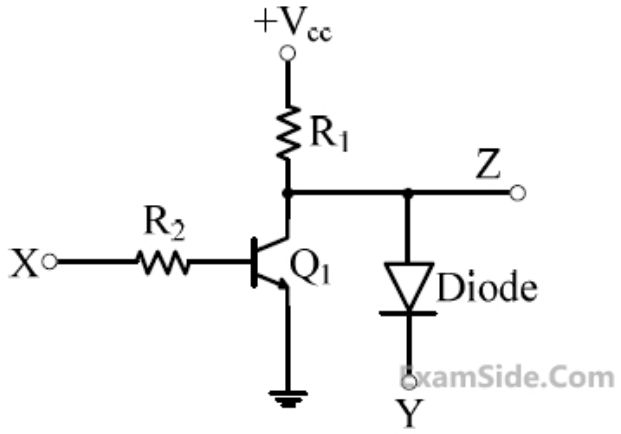
Fill in the blanks of the statements below concerning the following Logic Families:
Standard TTL (74XX), Low power TTL(74LXX) Low power schottky TTL(74LSXX), schottky TTL(74 SXXX), Emitter coupled Logic (ECL), CMOS
(a) Among the TTL Families, _____ family requires considerably less power than the standard TTL (74XX) and also has com parable proparation delay.
(b) Only the _____ family can operate over a wide range of power supply voltages.

Marks 2 6 questions

Q22. GATE ECE 2013

Open solution

In the circuit shown below, Q_1 has negligible collector-to-emitter saturation voltage and the diode drops negligible voltage across it under forward bias. If V_{CC} is +5 V, X and Y are digital signals with 0 V as logic 0 and V_{CC} as logic 1, then the Boolean expression for Z is

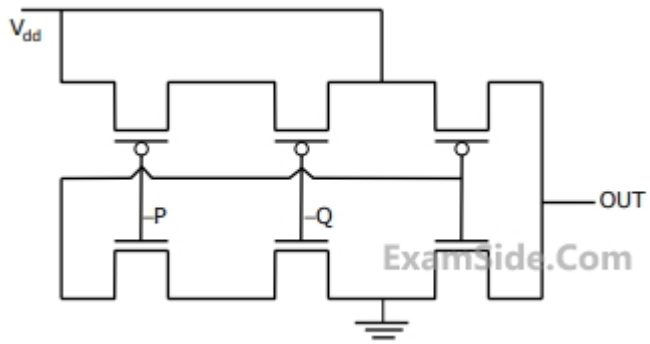


(A) XY (B) $\overline{X}Y$ (C) $X\overline{Y}$ (D) $\overline{XY}$

Q23. GATE ECE 2008

Open solution

The logic function implemented by the following circuit at the terminal OUT is



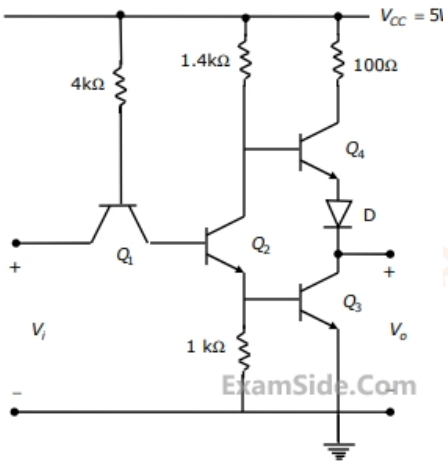
(A) P NOR Q (B) P NAND Q (C) P OR Q (D) P AND Q

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Q24. GATE ECE 2007

Open solution

The circuit diagram of a standard TTL NOT gate is shown in the figure. When V_i = 2.5V, the modes of operation of the transistors will be:



(A) Q_1 :reverse active
Q_2; normal active
Q_3; :saturation;
Q_4 :cut-off

(B) Q_1 :reverse active
Q_2: saturation
Q_3: saturation
Q_4 : cut-off

(C) Q_1 : normal active
Q_2; cut-off
Q_3; cut-off
Q_4 : saturation

(D) Q_1 : :saturation
Q_2: :saturation
Q_3 :saturation
Q_4 : normal active

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Q25. GATE ECE 2003

The DTL, TTL, ECL and CMOS families of digital ICs are compared in the following 4 columns

	(P)	(Q)	(R)	(S)
Fanout is minimum	DTL	DTL	TTL	CMOS
Power consumption is minimum	TTL	CMOS	ECL	DTL
Propagation delay is minimum	CMOS	ECL	TTL	TTL

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The correct column is

(A) P (B) Q (C) R (D) S

Open solution

Q26. GATE ECE 1994

In the output stage of a standard TTL, we have a diode between the emitter of the pull up transistor and the collector of the pull-down transistor. The purpose of this diode is to isolate the output node from the power supply V_{cc}.

(A) TRUE (B) FALSE

Open solution

Q27. GATE ECE 1987

Given that for a logic family,
V_{OH} is the minimum output high-level voltage
V_{OL} is the minimum output low-level voltage
V_{IH} is the minimum output high-level voltage and
V_{IL} is the minimum output low-level voltage.
The correct relationship is:

(A) $V_{IH} > V_{OH} > V_{IL} > V_{OL}$ (B) $V_{OH} > V_{IH} > V_{IL} > V_{OL}$

(C) $V_{IH} > V_{OH} > V_{OL} > V_{IL}$ (D) $V_{OH} > V_{IH} > V_{OL} > V_{IL}$

Open solution

Marks 8 1 questions

Q28. GATE ECE 1988

Implement the function $F = \left(\overline{A} + \overline{B}\right)\left(\overline{C} + \overline{D}\right)$ using two open collector TTL 2-input NAND gates.

Open solution

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Answer Key

Marks 1

1. **C**	2. **A**	3. **C**	4. **C**	5. **C**
6. **A**	7. **B**	8. **C**	9. **D**	10. **B**
11. **A**	12. **C**	13. **B**	14. **C**	15. **C**
16. **D**	17. $\overline{A+BC}$	18. **C(A+B)**	19. **A, C**	20. **D**
21. **(a) 74LS (a) CMOS**				

Marks 2

22. **B**	23. **D**	24. **B**	25. **B**	26. **B**
27. **B**				

Marks 8

28. **Solve It.**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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